

2nd Exam Spring 2021

٧/٧/٢٠٢١ ٥:٠٩ م

Math 102: Calculus II Second Exam

Monday, May 21, 2021

1. (20 pts.) Determine whether the following series is absolutely convergent, conditionally convergent, or divergent.

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt{n}}{n+1}$$

2. (20 pts.) Find the interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{5^n (x-2)^n}{n+1}$$

3. (15 pts.) Show that the sequence $a_n = \ln(n+1) - \ln n$ is bounded and monotonic.

4. (15 pts.) How many terms of the series

$$\sum_{n=1}^{\infty} (-1)^n \frac{1}{n10^n}$$

do you need to add to find the sum of the series correct to within an error of 10^{-4} ?

5. (15 pts.) Show that the series

$$\sum_{n=1}^{\infty} \frac{n \cos \frac{n\pi}{3}}{2^n}$$

is convergent.

6. (15 pts.) Find the Maclaurin series for

$$f(x) = \int \frac{\sin x^3}{x^2} dx$$

and its radius of convergence.